Behavior Poisoning in Cooperative MAPPO

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Key finding: Behavior poisoning was not reliably harmful; effects were seed-sensitive, mixed, and non-monotonic in p.

1. Motivation and Research Question

In cooperative MARL, training data is not independent: each agent's learning signal depends on trajectories produced jointly with teammates. A compromised teammate can alter the training distribution for the whole team without touching rewards, observations, gradients, or weights.

Research question

Can action-level behavior poisoning during MAPPO rollout collection degrade final clean-evaluation performance in `simple_spread`?

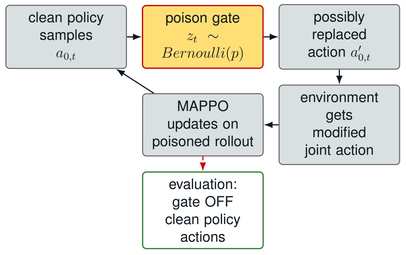
Why behavior poisoning is subtle	Why it matters here
Training-time only	Attack disappears at evaluation; any effect must be learned.
Low access	Attacker controls one agent's action, not the learner.
Team coupling	One bad teammate changes joint rollouts for all agents.

Task. In `simple_spread`, three agents cover three landmarks while avoiding unnecessary collisions. Good coordination spreads agents across distinct landmarks.

Core idea

training-time action replacement → changed rollout distribution → seed-dependent clean-evaluation outcome

2. Threat Model and Attack Mechanics



$$a'_{0,t} = \begin{cases} \text{poison}(a_{0,t}), & z_t = 1, \\ a_{0,t}, & z_t = 0. \end{cases}$$

Hook. Sample agent 0 action; sample z_t; replace if $z_t = 1$; step with modified joint action; update MAPPO. During evaluation set $z_t = 0$.

Can control	Agent 0 action during training rollout collection
Cannot control	observations, rewards, parameters, gradients, update rule, evaluation actions
Objective/result	objective is $D > 0$; observed D is not consistently positive

3. Attack Variants and Experimental Grid

Attack	Replacement behavior	What it tests
Random	uniform legal action	Does unstructured teammate noise shift learning?
Targeted no-op	force action 0	Does a passive teammate induce bad coordination?
KL-targeted	prefer no-op under KL budget	Can a constrained attack create positive D?

Experimental grid

5 clean baselines + 3 attacks $\times$ 3 poison probabilities $\times$ 5 seeds = 50 matched training runs. Each checkpoint uses 32 deterministic clean evaluation episodes; $p \in \{0.1, 0.2, 0.3\}$ and attack is disabled during evaluation. Matched seeds compare poisoned(seed s) against clean(seed s).

4. Metric and How to Read Degradation

Reward degradation

$$D_{seed,attack} = R_{clean,seed} - R_{poisoned,seed}$$

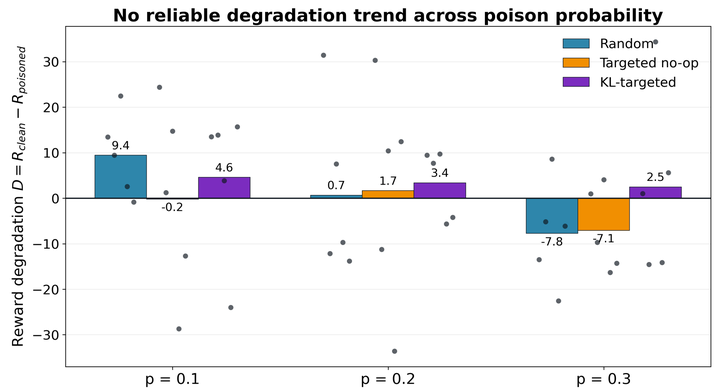
Value of D	Meaning
$D > 0$	poisoned policy evaluated worse than matched clean
$D = 0$	no matched-seed difference
$D < 0$	poisoned policy evaluated better than clean

Example

If $R_{clean} = -120$ and $R_{poisoned} = -135$, then $D = 15$ and the poisoned policy evaluated worse. If $R_{poisoned} = -110$, then $D = -10$ and the poisoned policy evaluated better.

Reward: $R_t \approx -\sum_{\ell \in L} \min_i \|x_i - x_\ell\|_2 - \lambda C_t$. Rewards are usually negative, so closer to 0 is better. Coverage distance and collision penalties both matter.

5. Main Result: Mixed Effects, Not Reliable Degradation



Caption. Positive D means the poisoned policy evaluated worse than the matched clean policy; negative D means it evaluated better. Random and no-op are not reliably harmful and become negative on average at $p = 0.3$.

p	Random	Targeted no-op	KL-targeted	Any $D > 0$	All $D > 0$
0.1	9.43	-0.21	4.58	5/5	3/5
0.2	0.65	1.66	3.39	4/5	2/5
0.3	-7.75	-7.06	2.47	3/5	1/5

Any $D > 0$ = at least one attack degrades that matched seed. All $D > 0$ = all three attacks degrade that matched seed.

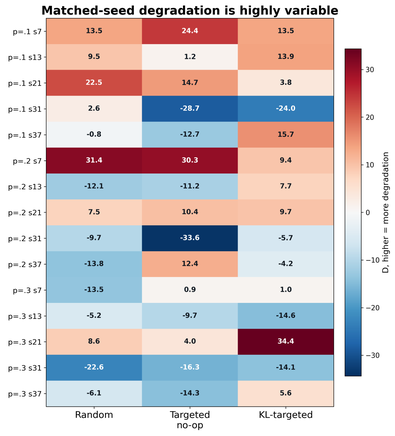
What the plot supports

- Some settings degrade: several attack/p averages have $D > 0$.
- Degradation is not reliable: random and no-op become beneficial on average at $p = 0.3$.
- Increasing p does not increase degradation.
- KL-targeted has positive mean D at all tested p, but not stable per-seed outcomes.

Answer and caution

Answer: sometimes, but not reliably. Positive examples exist, but many matched settings show no degradation or improvement. The data support mixed outcomes, not stable attack success.

6. Seed Sensitivity

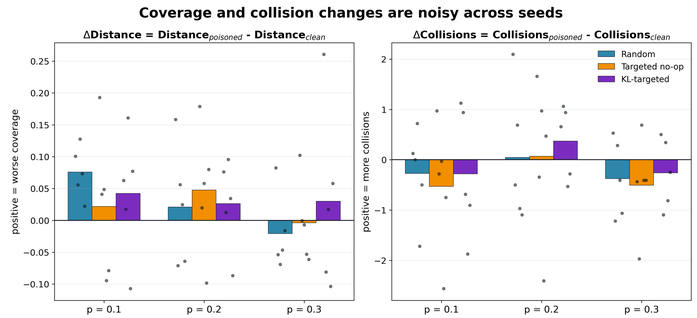


Caption. Each cell is matched-seed reward degradation. Red cells indicate degradation; blue cells indicate improvement over clean. The mixture shows highly seed-dependent outcomes.

Seed-level lesson

Seed-level variation is the central observation: a single clean/poisoned comparison could overstate or understate the attack. Several negative cells mean stable degradation is not shown.

7. Mechanism: Coverage and Collision Effects Are Noisy



Caption. Coverage and collision changes overlap across attacks and p values. These plots do not reveal a simple mechanism that consistently explains reward degradation.

- Reward includes both landmark coverage and collision penalties.
- Some poisoned policies improve one component while worsening another.
- Overlap is consistent with seed- and trajectory-dependent outcomes, but does not prove a mechanism.

Future defense ideas

If future work finds stronger behavior-poisoning attacks, likely defenses would monitor rollout behavior rather than only rewards or gradients.

- Action-distribution anomaly detection.
- Per-agent entropy or intervention monitoring.
- Robust rollout filtering and adversarial teammate training.

8. Hypothesis Check

Hypothesis	Outcome	Evidence
H1 Clean MAPPO learns coordination	Baseline established	clean runs provide matched baselines
H2 Poisoning reliably degrades clean evaluation	Not supported	many settings have $D \leq 0$; random/no-op are negative at $p = 0.3$
H3 No-op beats random	Not supported	no-op does not reliably produce higher D
H4 KL-constrained poisoning reliably degrades	Weak/mixed support	positive mean D at all tested p, but variable by seed
H5 More poison means more degradation	Not supported	average degradation decreases or reverses for random/no-op

Bottom line

Supported: non-monotonicity and seed sensitivity. Partially supported: some settings degrade. Not supported: reliable degradation, targeted dominance, or monotonic harm.

9. Takeaways, Limitations, and Next Steps

Takeaways

- Action-level poisoning is a plausible training-time threat model, but this sweep did not show reliable degradation.
- Some matched-seed settings degrade clean-evaluation reward; many show no degradation or improvement.
- The most robust finding is non-monotonic, seed-sensitive behavior rather than stable attack success.
- Random and no-op poisoning improve on average at $p = 0.3$, suggesting possible exploration or regularization.
- KL-targeted has positive mean degradation at all tested p, but the heatmap shows seed dependence.

Limitations: one environment; one compromised agent; three p values; one no-op target action; five seeds; no confidence intervals; no adaptive target search. Conclusions are exploratory.

Next steps: add more seeds and confidence intervals before claiming reliable degradation; search target actions; test harder environments; compare against benign noise; check whether KL averages persist.

Main lesson

The attack model is plausible, but observed degradation is intermittent rather than reliable.

Refs: Yu et al. 2022; Schulman et al. 2017; Lowe et al. 2017; Terry et al. 2021; Mordatch and Abbeel 2018.